Straight-line motion with constant acceleration

Explain acceleration as the change in velocity each second. Find velocity and displacement for constant acceleration.

Understand the idea first

• A bicycle starts moving and gains speed.

• If its velocity changes by the same amount each second, its acceleration is constant.

Acceleration is not the velocity itself. It describes how velocity changes over time.

Kazakh terms in plain language

Қазақша	Тілдегі баламасы	Қарапайым түсінігі
үдеу	acceleration	How much velocity changes in each unit of time.
бастапқы жылдамдық	initial velocity	The velocity at the start of the motion being studied.
орын ауыстыру	displacement	The change from starting to ending position, including direction.

Status: physics and original answer key checked; translations and terms need teacher review.

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Quantities, formula, and example

v = v_0 + at

a = (v - v_0)/t

When to use it

Distance relates to the area under a v-t graph.

For constant acceleration, s = v□t + at²/2.

Displacement magnitude equals distance while direction does not reverse.

Symbols and units

Шама	Мағынасы
v□	initial velocity, m/s
v	velocity after time t, m/s
a	acceleration, m/s²
t	time, s
s	displacement along the chosen axis, m
v□	initial velocity, m/s
a	constant acceleration component, m/s²
t	time, s

Worked example

• Given: v□ = 2 m/s, a = 1.5 m/s², t = 4 s.

• v = 2 + 1.5·4 = 8 m/s.

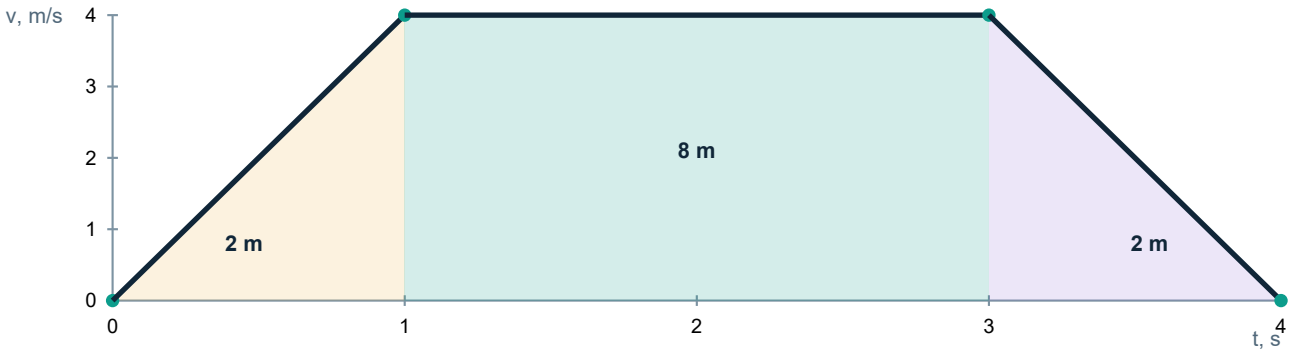
• s = 2·4 + 1.5·4²/2 = 20 m.

• Average velocity is 5 m/s; over 4 s that gives 20 m.

Graph question from an ENT collection

The velocity-time graph of a material point is shown. Find the distance travelled in 4 s.

Find the distance travelled in 4 seconds.



A) 13 m • B) 12 m • C) 9 m • D) 15 m

Solution and check

Distance is the sum of the areas under the graph: 2 m + 8 m + 2 m = 12 m. The original answer table (PDF p. 37) confirms choice B.

Common mistakes

• Do not treat acceleration as velocity: acceleration describes how velocity changes over time.

• Use s = v□t + at²/2 for constant acceleration; s is displacement along the chosen axis.

• For the graph question, do not use the velocity at t = 4 s as the distance: distance is the area under the graph.

Sources

Физика Закирова 10 ЖМ.pdf, §3, PDF pp. 15-18; textbook exercises: p. 20.

Testcenter -3.pdf, task 1: PDF p. 2; original answer table: p. 37. Choice B was checked by graph area.

Current NCT physics specification, applies from 2026

Status: physics and original answer key checked; translations and terms need teacher review.